PART A: Answer any **two** question

A-1. A mass at the end of a spring oscillates with an amplitude of 5 cm at a frequency of 1 Hz. At $t = 0$ the mass is at its equilibrium position ($x = 0$). [5]

(a) Find all possible equations describing the position of the mass as a function of time, in the form $x = A\cos(\omega t + \alpha)$, giving the numerical values of A, ω, and α. [2]

(b) What are the values of x, $\frac{dx}{dt}$, and $\frac{d^2x}{dt^2}$ at $t = \frac{8}{3}$ s? [3]

A-2. A cylinder of density ρ_c is floating vertically in a liquid of density $2\rho_c$. The cylinder has a length l and radius r. It oscillates vertically if it is tapped slightly. [5]

(a) Write the equation of motion. [2]

(b) Find the length of the cylinder above the liquid surface when it is in equilibrium. [2]

(c) What is the frequency of the oscillation? [1]

A-3. Consider two waves of angular frequencies, ω_1 and ω_2, and wavenumbers k_1 and k_2, respectively. [5]

(a) Calculate the phase and group velocities. [2]

(b) Can we have group velocity more than the phase velocity? What is the condition required? [3]

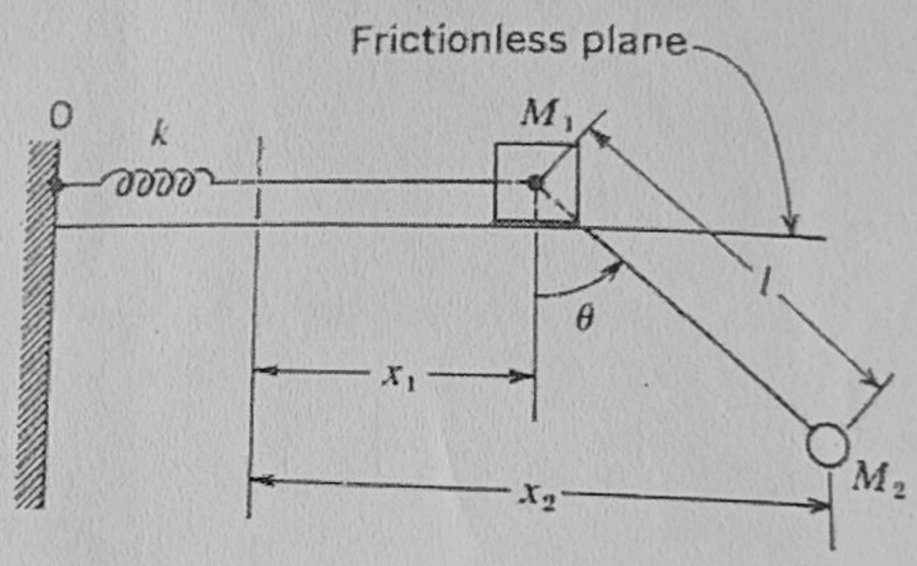
PART B: Answer any **five** questions

B-1. The sketch shows a mass m on a frictionless plane connected to support O by a spring of stiffness k. Mass M_2 is supported by a string of length l from M_1. [8]

(a) Using the approximation of small-angle [2

$$\sin\theta \approx \tan\theta \approx (x_2 - x_1)/l$$

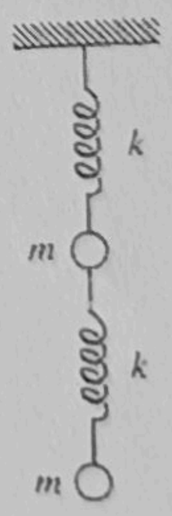
and starting from $F = ma$ derive the equation for motion of M_1 and M_2.



(b) For $M_1 = M_2 = M$ use the equations to obtain the normal frequencies of the system. [3

(c) What are the normal mode motions for $M_1 = M_2 = M$ $g/l \gg k/m$? [3

B-2. Two equal masses are connected as shown, with two identical massless strings of spring constant k. [8]



(a) Write the equations of motion. [2

(b) Considering only motion in the vertical direction, show that the angular frequency of the two normal modes are given by $\omega^2 = (3 \pm \sqrt{5})\frac{k}{2m}$ and hence that the ratio of the normal mode frequency is $(\sqrt{5}+1)/(\sqrt{5}-1)$. [4

(c) Find the ratio of the amplitude of the two masses in each separate mode. [2

B-3. (a) Write Maxwell's equations. [2]

(b) Write the electric field of a plane EM wave traveling along x direction. [1]

(c) Show that the electric field is transverse to the direction of propagation. [2]

(d) Show that the magnetic field is in the yz plane and is perpendicular to the electric field. [3]

B-4. Consider an electromagnetic wave incident at a boundary with non-zero angle of incidents. [8]

(a) What are the Fresnel equations for *s-type* and *p-type* waves? [2]

(b) Show that for an *s-type* wave there no angle for which the incident wave is completely transmitted (for *p-type* waves it is known as the *Brewster's angle*). [3]

(c) Show that at Brewster's angle for *p-type* polarized light, we have $\sin 2\theta_I = \sin 2\theta_T$. Is that a contradiction? Argue. [2+1]

B-5. Consider Young's double slit experiment, where lights from two thin slits, kept at a separation of d distance, create interference pattern on a monitor at a distance D. Assuming that small angle approximation holds and the entire setup is kept in air (refractive index 1), answer the following. [8]

(a) Express the intensity of interference pattern as a function of the distance x from the center of the monitor. [2]

(b) What is the distance between successive minima? [2]

(c) If we place a thin slab of glass of thickness $\delta\ell$ and refractive index η in front of S_2, calculate the resulting intensity pattern. [4]

B-6. The refractive index of the glass is 3/2. Consider a glass-air interface and a linearly polarized *p-type* wave is incident on the interface. [8]

(a) Calculate the angle for total internal reflection. [2]

(b) Calculate the Brewster's angle. [2]

(c) What is the angle at which the reflected and the transmitted beams have the same intensities? [4]

B-7. Consider a a Young's double slit interferrometer and a monochromatic source with wavelength λ. d is the distance between the slits. Add another two slits at a uniform distance d to create a linear four slit interferrometer. Assume the slits to be thin. [8]

(a) Calculate the amplitude of the waves from the four slits as they arrive at a place on the screen D distance away (as it happens in a double slit experiments). [4]

(b) Calculate the resulting intensity pattern as a function of distance x on the screen from the central maximum or central minimum. Use small angle approximations, i.e., $\sin\theta \sim \tan\theta \sim \theta$. [2]

(c) Sketch qualitatively the intensity as a function of x. Compare it with the interference pattern of the double-slit case by plotting it on the same graph. [2]